

Project Performance Audit & Cost Report

Generated on July 09, 2026 | Enterprise Sourcing Live Feed

Executive Summary

This is a Commercial building with 5 structural divisions, including 2 bedrooms, 2 bathrooms, 1 kitchen, and 1 hall. Based on the uploaded blueprint, the estimated construction cost is ₹13,54,85,225 using Luxury quality specifications. The estimated completion time is 42 months.

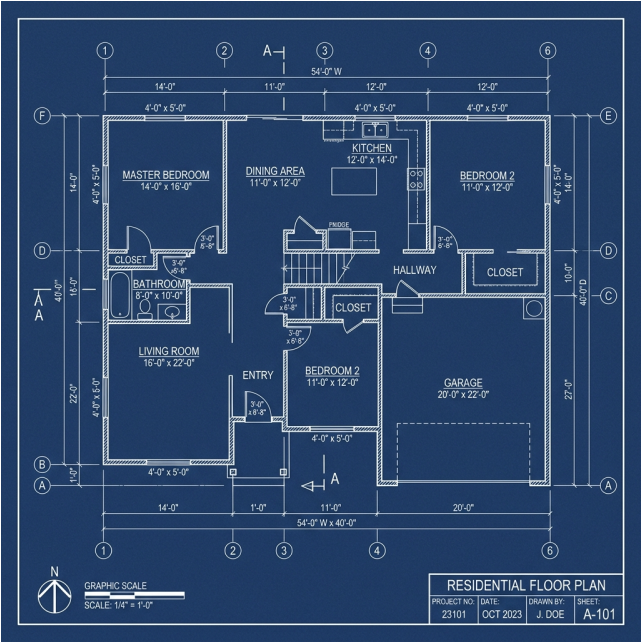
Project Details & Assigned Personnel

Project Name:	Downtown Heights Plaza	Client Name:	Municipal Development Corp
Location:	Connaught Place, New Delhi	Building Type:	Commercial
Floors:	15	Material Quality:	Luxury
Assigned PM:	Marcus Aurelius	Assigned Engineer:	Diana Prince

AI Blueprint Layout Analysis

Structural Division	Detected Quantity	Classification
Bedrooms / Cabin Areas	2 Zones	Enclosure Space
Bathrooms / Plumbing	2 Zones	Sanitary Block
Kitchen / Cafeteria	1 Zone	Services Block
Hall / Reception Zone	1 Zone	Common Area
Dining / Pantry Area	1 Zone	Amenity space
Balconies	15 Zones	Open Air Extension
Parking Slots	1 Zone	Utility
Staircase / Lift Core	1 Zone	Vertical Circulation
Doors Identified	60 Units	Apertures / Access
Windows Identified	90 Units	Apertures / Ventilation
Total Built-up Area	18,000 sqft	Calculated Area

Project Blueprint Layout Drawing Reference:



AI Material Quantity & Sourcing Schedule

Material Item	Estimated Qty	Unit	Total Cost (INR)
Cement	7,200	bags	₹34,56,000
Steel Rebar	72.0	tons	₹59,04,000
Bricks	396,000	pcs	₹47,52,000
Sand	1,440	tons	₹25,92,000
Blue Metal Aggregate	1,080	tons	₹21,60,000
Gravel / Soling	720	tons	₹10,08,000
Ready-Mix Concrete	540	cum	₹35,10,000
Paint	2,880	liters	₹14,97,600
Flooring Tiles	20,700	sqft	₹53,82,000
Electrical Conduit & Wiring	1	lump-sum	₹57,60,000
Plumbing Pipes & Fixtures	1	lump-sum	₹46,80,000
Roofing/Slab Concrete reinforcement	1	lump-sum	₹75,60,000
Doors	60	units	₹15,00,000
Windows	90	units	₹16,20,000
Labor Sourcing Requirement	15,300	man-days	₹1,45,35,000

Financial Summary & Cost Breakdown

Cost Category	Estimated Cost (INR)	Cost Category	Estimated Cost (INR)
Materials Base Cost	₹5,13,81,600	Labor Cost	₹1,45,35,000
Equipment & Machinery	₹1,08,00,000	Electrical Works	₹57,60,000
Plumbing & Sewerage	₹46,80,000	Painting & Coatings	₹14,97,600
Roofing Works	₹75,60,000	Finishing Works	₹99,99,600
Government Approvals	₹5,00,000	Miscellaneous Expenses	₹26,36,664
Contingency Amount (5%)	₹54,67,523	GST (18% applicable)	₹2,06,67,238
Grand Total Estimate	₹13,54,85,225		

Construction Timeline & Project Schedule

Construction Phase / Stage	Estimated Duration (Weeks)
Site Preparation	12
Foundation	30
Column Work	23
Wall Construction	27
Roofing	18
Electrical Work	15
Plumbing	12
Flooring	17
Painting	14
Finishing	18
Overall Estimated Duration	42 Months

AI Construction Risk Scorecard

Risk category	Level	Telemetry Diagnosis	Mitigation Recommendation
Delay Risk	Medium	Slight logistical supply delay risks or approval buffer slip.	Maintain a 14-day schedule buffer.
Budget Risk	Medium	Market steel price volatility.	Procure major materials at project kickoff.
Safety Risk	High	Scaffolding falls and heavy vehicle movements.	Enforce mandatory hard-hats and harness usage.
Material Sourcing	Medium	Sourcing delays for premium finishes.	Source from multiple supplier agreements.
Weather Risk	Low	Seasonal monsoon rainfall interruptions.	Perform roofing concrete pours before peak rainy season.

AI Sourcing & Value Optimization Recommendations

- Suggested Workforce: 144 active workers (skilled & unskilled)
- Suggested Machinery: Concrete Mixer Machine, Slab Vibrator, JCB Excavator, Tower Crane, Passenger Hoist Lift
- Cost Optimization: Optimize cut-lists for steel rebar to minimize waste below 5%.
- Construction Sequence: Run electrical conduiting concurrently with brick laying partitions.
- Material Sourcing Improvements: Use fly-ash blended OPC cement for columns foundation slab to lower footprint.
- Safety Program: Conduct safety toolboxes each morning for working-at-height laborers.

Final Conclusion: This cost and quantity report is generated based on standard mechanical layouts and area bounding boxes detected from the building blueprint. The quantities and rates comply with current Central Public Works Department (CPWD) schedule indices in India and are ready for commercial execution review.

Marcus Aurelius
Project Manager

Sarah Administrator
Auditor Coordinator